

HARSHITHAN KAVITHA SUKUMAR

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EDUCATION

UNIVERSITY OF MINNESOTA, Carlson School of Management, MN

Master of Science in Business Analytics - Artificial Intelligence for Business track

August 2025

ANNA UNIVERSITY, Chennai, India

Bachelor of Technology in Information Technology

June 2024

EXPERIENCE

CARLSON ANALYTICS LAB, Minneapolis, MN

Data Science Student Consultant (Tech Lead)

August 2024 – Present

Agentic Chatbot Performance Analytics in Education | Python, SQL, PyTorch, RAG, Azure, MySQL, Pinecone Vector Database

- Led client engagement to architect an enterprise AI query assistant for 100,000+ users through a hybrid RAG design with intelligent routing and diagnostics. Audited model and retrieval performance end-to-end to surface bottlenecks impacting resolution rate, latency, and self-serve deflection.
- Analyzed 200,000 user queries and taxonomy gaps to expand intents from 9 to 25+. Fine-tuned DistilBERT in PyTorch, lifting F1 by 36% versus the production baseline and improving precision on top intents tied to CSAT and containment..
- Built ELT pipelines on Azure and MySQL with SQL and Python to track accuracy, latency, cost, and resolution. Added retrieval-trace diagnostics, error taxonomy, and cohort/funnel analyses to link failure modes to KPI shifts and drive backlog.
- Conducted cost and latency trade-off studies and implemented a quantization-based inference stack with privacy controls. Cut cloud spend by 70% while maintaining 98% of model accuracy and reducing API overhead by 60% against SLA targets.

Strategic Consulting Engagement: NCAA Policy Impact Assessment | Python, Excel, Causal Inference, A/B Testing, EDA, PSM

- Analyzed 196,000+ NCAA and U Sports player-week records to quantify the causal effect of NIL policy on performance across a multi-year panel. Framed it as a quasi-experiment and built treatment–control cohorts tied to on-field KPIs.
- Applied propensity score matching and difference-in-differences with a Canadian control group to estimate effect sizes and confidence intervals. Controlled for team and season heterogeneity with fixed effects and clustered standard errors.
- Designed a scalable Python pipeline with distributed processing for feature engineering, time-series aggregation, and EDA across 15+ variables. Added data validation and reproducible notebooks to support auditability and rapid iteration.
- Consulted athletic department executives on \$2M+ resource allocation using scenario modeling and sensitivity analyses. Recommendations informed policy decisions across 10+ client organizations and set a monitoring plan for KPIs and tests.

HEXAMAP SOLUTIONS INDIA PVT, Chennai, India

Data Analyst Intern

January 2024 - May 2024

- Developed an optimization model for cell tower placement using Excel, SQL, Python, QGIS, and Tableau, piloted across 24 wards. Achieved ~20% potential coverage improvement to guide network rollout.
- Designed executive-level Tableau dashboards with dynamic KPI tracking and drill-downs. Enabled real-time, data-driven decisions for 10+ project leads in network optimization.
- Integrated geospatial analysis, statistical modeling, and visualization to connect infrastructure spend with measurable service gains. Provided a scalable framework for future expansion

PROJECTS

Wholesale Customer Segmentation: B2B Distribution | R, K-Means, Hierarchical Clustering, dplyr, correlation analysis

- Built Segmented 440 customers into three behavior based clusters with a silhouette score around 0.44 and surfaced a retail heavy segment of more than 40% of annual spend, informing targeted campaigns, inventory set points, and route planning.

Market Mix Modeling For Media ROI Optimization | Python, Nested CV, Regression Models, LightGBM, Mlflow

- Delivered production-grade MMM solution to attribute revenue across six paid channels, driving 20–35% projected ROI lift through adstock-based modeling, automated budget reallocation, and real-time diagnostics aligned with post-iOS14 strategy.

Bankiq: Banking Simulation For Risk & Customer Analytics | Azure, SQL, DBT, Great Expectations, Random Forest

- Engineered a banking simulation across five domains, enabling fraud, churn, and segmentation modeling with 96.7% validation accuracy and audit traceability. Delivered ML pipelines processing 1M+ transactions using scalable workflows.

Streaming Stock Sentiment Prediction | PySpark, Databricks, Structured Streaming, Logistic Regression, TF-IDF, NLP

- Designed a real-time sentiment classification pipeline for financial tweets using Spark Streaming, enabling sub-second signal detection and anomaly monitoring. Achieved 85% accuracy, supporting faster decision-making in trading environments.

TECHNICAL SKILLS

- **Languages & libraries:** Python, R, SQL, Java, JavaScript..
- **Data Platforms & Analytics Tools:** PyTorch, Langchain, Azure, Snowflake, Databricks, AWS, MySQL, Tableau, Power BI, Excel, Jira, Git, Scikit-learn, dbt, Airflow, Apache PySpark, Kafka, Looker, MLflow.
- **Techniques & Concepts:** EDA, Segmentation, clustering, Metric design, KPI frameworks, Regression, Boosting Models, Tree-Based Models, NLP, Generative AI, Causal Inference, Time Series Forecasting, Recommender Systems, A/B Testing.